

Random Walks on Graphs

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1 Venn Diagram of Class of Markov Chains, Examples of Each Region:

We know that irreducible Markov chains are either positive recurrent or null recurrent or transient. Also in a finite Markov chain there exist at least one positive recurrent state and hence a finite Markov chain can't be null recurrent or transient. Also an finite irreducible Markov chain is always positive recurrent. Combining this all we make a Venn-diagram for class of Markov chains. Figure-5 represents the Venn diagram of class of Markov chains.

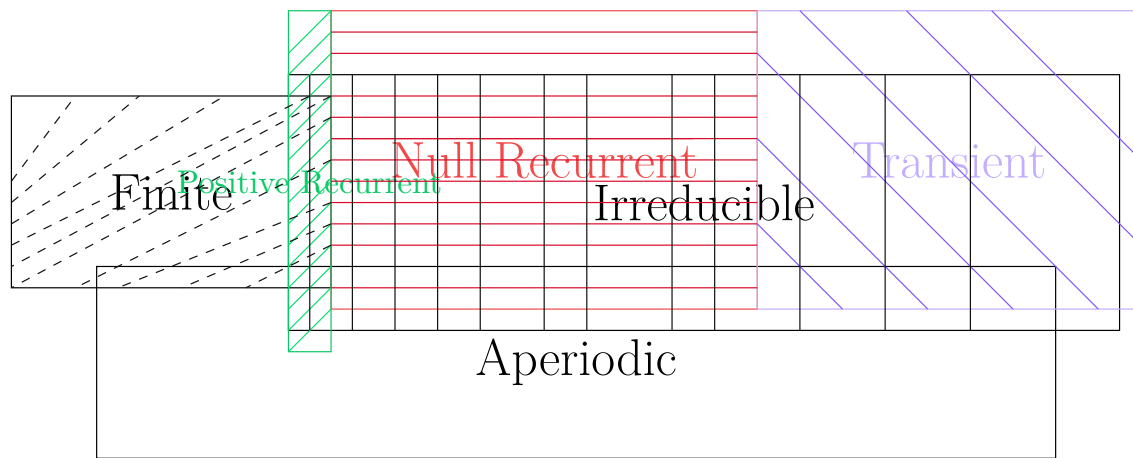


Figure 1: Venn Diagram

1.1 Example of Finite, Positive-recurrent, Aperiodic, Irreducible:

Consider a Markov chain with state-space $\{0, 1, 2, 3\}$ satisfying $P(0, 0) = P(3, 3) = \frac{1}{2}$, $P(0, 1) = P(1, 2) = P(2, 3) = P(3, 2) = P(2, 1) = P(1, 0) = \frac{1}{2}$.

Observe that state-space is finite, chain is irreducible as from each state to go to any other state, we have positive probability, chain is aperiodic as it is irreducible with states 0, 3 having period 1 and chain is positive-recurrent because in finite chain at least one state is positive recurrent hence the chain being irreducible all states are positive-recurrent.

1.2 Example of Finite, Positive-recurrent, Aperiodic, Reducible:

Consider a Markov chain with state-space $\{0, 1, 2, 3\}$ satisfying $P(0, 1) = P(3, 3) = P(2, 0) = 1, P(1, 2) = P(1, 0) = \frac{1}{2}$.

Observe that the state-space is finite, there are two disjoint communicating classes $\{0, 1, 2\}$ and $\{3\}$ which means the chain is reducible, chain is aperiodic as each state has period 1, chain is positive recurrent as $P(3, 3) = 1$ and in $\{0, 1, 2\}$ (which is a communicating class) there must exist a state which is positive-recurrent which implies all states are positive-recurrent.

1.3 Example of Infinite, Positive-recurrent, Aperiodic, Irreducible:

Consider a Birth-Death chain on the non-negative integers. Then the transition

function is $P(x, y) = \begin{cases} q_x & y = (x - 1) \\ r_x & y = x \\ p_x & y = (x + 1) \end{cases}$, where $(p_x + q_x + r_x) = 1$ and for

all $x \in \mathbb{N} \cup \{0\}$, $p_x, q_x, r_x > 0$. If $\sum_{y=0}^{\infty} \frac{p_0 p_1 \dots p(y-1)}{q_1 q_2 \dots q_y} < \infty$, then it is an Infinite, Positive-recurrent, Aperiodic, Irreducible Markov chain.

1.4 Example of Infinite, Positive-recurrent, non-Aperiodic, Irreducible:

Consider a Birth-Death chain on the non-negative integers. Then the transition

function is $P(x, y) = \begin{cases} q_x & y = (x - 1) \\ r_x & y = x \\ p_x & y = (x + 1) \end{cases}$, where $(p_x + q_x + r_x) = 1$ and for

all $x \in \mathbb{N} \cup \{0\}$, $p_x, q_x, r_x > 0$. If $\sum_{y=0}^{\infty} \frac{p_0 p_1 \dots p(y-1)}{q_1 q_2 \dots q_y} < \infty$, then it is a Infinite, Positive-recurrent, non-Aperiodic, Irreducible Markov chain.

1.5 Example of Infinite, Transient, Aperiodic, Irreducible:

Consider a Birth-Death chain on the non-negative integers. Then the transition

$$\text{function is } P(x, y) = \begin{cases} q_x & y = (x - 1) \\ r_x & y = x \\ p_x & y = (x + 1) \end{cases}, \text{ where } (p_x + q_x + r_x) = 1 \text{ and for all}$$

$x \in \mathbb{N} \cup \{0\}$, $p_x, q_x, r_x > 0$. If $\sum_{y=0}^{\infty} \frac{q_1 q_2 \dots q_y}{p_1 p_2 \dots p_y} < \infty$ then it is an Infinite, Transient, Aperiodic, Irreducible Markov chain.

1.6 Example of Infinite, Transient, non-Aperiodic, Irreducible:

Consider a Birth-Death chain on the non-negative integers. Then the transition

$$\text{function is } P(x, y) = \begin{cases} q_x & y = (x - 1) \\ r_x & y = x \\ p_x & y = (x + 1) \end{cases}, \text{ where } (p_x + q_x + r_x) = 1 \text{ and for all}$$

$x \in \mathbb{N} \cup \{0\}$, $p_x, q_x, r_x > 0$. If $\sum_{y=0}^{\infty} \frac{q_1 q_2 \dots q_y}{p_1 p_2 \dots p_y} < \infty$ then it is an Infinite, Transient, non-Aperiodic, Irreducible Markov chain.

1.7 Example of Infinite, Null-recurrent, Aperiodic, Irreducible:

Consider a Birth-Death chain on the non-negative integers. Then the transition

$$\text{function is } P(x, y) = \begin{cases} q_x & y = (x - 1) \\ r_x & y = x \\ p_x & y = (x + 1) \end{cases}, \text{ where } (p_x + q_x + r_x) = 1 \text{ and for all}$$

$x \in \mathbb{N} \cup \{0\}$, $p_x, q_x, r_x > 0$. If both $\sum_{y=0}^{\infty} \frac{q_1 q_2 \dots q_y}{p_1 p_2 \dots p_y} = \infty$, $\sum_{y=0}^{\infty} \frac{p_0 p_1 \dots p_{(y-1)}}{q_1 q_2 \dots q_y} = \infty$ then it is an Infinite, Null-recurrent, Aperiodic, Irreducible Markov chain.

1.8 Example of Infinite, Null-recurrent, non-Aperiodic, Irreducible:

Consider a Birth-Death chain on the non-negative integers. Then the transition

$$\text{function is } P(x, y) = \begin{cases} q_x & y = (x - 1) \\ r_x & y = x \\ p_x & y = (x + 1) \end{cases}, \text{ where } (p_x + q_x + r_x) = 1 \text{ and for all}$$

$x \in \mathbb{N} \cup \{0\}$, $p_x, q_x, r_x > 0$. If both $\sum_{y=0}^{\infty} \frac{q_1 q_2 \dots q_y}{p_1 p_2 \dots p_y} = \infty$, $\sum_{y=0}^{\infty} \frac{p_0 p_1 \dots p_{y-1}}{q_1 q_2 \dots q_y} = \infty$ then it is an Infinite, Null-recurrent, non-Aperiodic, Irreducible Markov chain.

1.9 Example of Infinite, Reducible, Positive-recurrent:

A Markov chain with more than one positive-recurrent, infinite disjoint communicating classes is such an example.

1.10 Example of Infinite, Reducible, Null-recurrent:

Consider a Markov chain on non-negative integers with transition matrix $P(2i, 2i+2) = P(2i, 2i-2) = P(2i+1, 2i+3) = P(2i+1, 2i-1) = \frac{1}{2} \forall i \in \mathbb{N}$, $P(0, 2) = P(1, 3) = P(0, 0) = P(1, 1) = \frac{1}{2}$.

Observe that it is disjoint union of two irreducible state-space, i.e; χ isn't irreducible.

Also observe, $P^{(2n)}(0, 0) = \frac{(2n)!}{(n!)^2 2^{2n}} \approx \frac{1}{\sqrt{\pi n}}$ and $P^{(2n)}(1, 1) = \frac{(2n)!}{(n!)^2 2^{2n}} \approx \frac{1}{\sqrt{\pi n}}$.

Now $\sum_{n \in \text{even}} P^{(2n)}(0, 0) \approx \sum_{n \in \text{even}} \frac{1}{\sqrt{\pi n}}$ and $\sum_{n \in \text{odd}} P^{(2n)}(1, 1) \approx \sum_{n \in \text{odd}} \frac{1}{\sqrt{\pi n}}$

Now, say $\sum_{n \in \text{even}} P^{(2n)}(0, 0) = S$ and $\sum_{n \in \text{odd}} P^{(2n)}(1, 1) = A$.

Then

$$\begin{aligned} (S + A) &\approx \sum_{i=1}^{\infty} \frac{1}{\sqrt{\pi n}} \\ \Rightarrow \frac{\sum_{i=1}^{\infty} \frac{1}{\sqrt{\pi n}}}{\sqrt{2}} + A &\approx \sum_{i=1}^{\infty} \frac{1}{\sqrt{\pi n}} \quad (\text{as } \sum_{n \in \text{even}} \frac{1}{\sqrt{\pi n}} = \frac{\sum_{i=1}^{\infty} \frac{1}{\sqrt{\pi n}}}{\sqrt{2}}) \\ \Rightarrow A &\approx (1 - \frac{1}{\sqrt{2}}) \sum_{i=1}^{\infty} \frac{1}{\sqrt{\pi n}} = \infty \quad (\text{as we know } \sum_{i=1}^{\infty} \frac{1}{\sqrt{n}} = \infty) \end{aligned}$$

Also, $S \approx \infty$ as $\sum_{i=1}^{\infty} \frac{1}{\sqrt{n}}$. Hence 0 and 1 are recurrent. As by construction the chain has two communicating classes: one with all odd states and another with all even states, 0 is in one of the communicating class making the class recurrent and 1 is in another making that recurrent. So, the Markov chain is recurrent.

Now $\sum_{n \in \text{even}} 2n P^{(2n)}(0, 0) = 2 \sum_{n \in \text{even}} \sqrt{\frac{n}{\pi}} = \infty$ and $\sum_{n \in \text{odd}} 2n P^{(2n)}(1, 1) = 2 \sum_{n \in \text{odd}} \sqrt{\frac{n}{\pi}} = \infty$ which implies 0 and 1 are null-recurrent, hence the Markov chain is null-recurrent also.

1.11 Example of Infinite, Reducible, Transient:

Suppose customers are arriving at a service-facility at various times and are eventually served. If there are customers arrived but not yet served, a queue is formed. The queue is observed after regular time-periods, say a minute. If there are any customer waiting in the beginning of a minute, exactly one of them will be served in the next minute.

Let the no. of people arriving in the n th minute be $\zeta_n, \forall n \geq 1$. $\zeta_1, \zeta_2, \dots$ are iid having a discrete distribution g . Let X_n be the no. of people in the queue after the n th minute. Let X_0 be initial no. of people in the queue.

If $X_n = 0$, then $X_{n+1} = \zeta_{n+1}$ and if $X_n \geq 1$, then $X_{n+1} = (X_n + \zeta_{n+1} - 1)$. Here, $p_{0j} = P(X_1 = j | X_0 = 0) = P(\zeta_1 = j | X_0 = 0) = g(j)$, where $j \in \mathbf{N} \cup \{0\}$ and for $i > 0, p_{ij} = P(X_1 = j | X_0 = i) = P(\zeta_1 = j - i + 1 | X_0 = i) = g(j - i + 1)$.

The transition matrix is:

$$P = \begin{pmatrix} g(0) & g(1) & g(2) & g(3) & g(4) & \dots \\ g(0) & g(1) & g(2) & g(3) & g(4) & \dots \\ 0 & g(0) & g(1) & g(2) & g(3) & \dots \\ 0 & 0 & g(0) & g(1) & g(2) & \dots \\ 0 & 0 & 0 & g(0) & g(1) & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

If $g(0) = 0$, then P becomes upper-triangular matrix and as multiplying two upper-triangular matrices we get again a upper-triangular matrix, $\forall n \in \mathbf{N}, P^n$ remains upper-triangular. So there does not exist n such that $P^{(n)}(i, j) > 0, \forall j < i$. Hence it is not irreducible. Also as there does not exist n such that $P^{(n)}(i, j) > 0, \forall j < i$, once we reach $j > i$ from any i , we can never again return to i , which means all states are transient making the Markov chain transient.

2 Recurrence and Transience of Random Walk On $\mathbb{Z}^d$ Lattices:

We want to check whether the chain is recurrent or transient on lattices of integer dimensions.

2.1 $d = 1$ (Irreducible state space of all integers:)

The transition probabilities are defined by:

1. $\mathbb{P}(X_{i+1} = k + 1 \mid X_i = k) = p$
2. $\mathbb{P}(X_{i+1} = k - 1 \mid X_i = k) = 1 - p = q$

Note that $0 \rightarrow 0$ in odd steps is not possible. (So the chain has period 2.)

Now, In $2n$ steps:

$$p_{00}^{(2n)} = \binom{2n}{n} p^n q^n$$

$$\sum_{n \in \mathbb{N}} p_{00} = \sum_{n=1}^{\infty} \binom{2n}{n} (pq)^n$$

$$\sim \sum_n \frac{(4pq)^n}{\sqrt{n}}$$

If $p = q = \frac{1}{2}$, then $4pq = 1$, and the sum

$$\sum \frac{(4pq)^n}{\sqrt{n}} = \sum \frac{1}{\sqrt{n}} \text{ diverges, and the chain becomes recurrent.}$$

Else if $4pq < 1$, we can bound it by a geometric series, hence the sum converges. Therefore the chain is transient.

So, in other words, simple symmetric random walk in $d = 1$ is recurrent while simple asymmetric random walk in $d = 1$ is transient.

2.2 $d = 2$

The state-space S is,

$$S = \{(i, j) \mid i \in \mathbb{Z}, j \in \mathbb{Z}\}$$

Now we can move to right, left, up and down with equal probability $\frac{1}{4}$. So, we have

$$P_{(0,0)(0,0)}^{(2n)} = \sum_{i=0}^n \sum_{j=0}^n \frac{(2n)!}{i! j! (n-i)! (n-j)!} \left(\frac{1}{4}\right)^{2n}$$

Hence we have,

$$\begin{aligned} & \sum_{i=0}^n \sum_{j=0}^n \frac{(2n)!}{n! n!} \cdot \frac{n!}{i! (n-i)!} \cdot \frac{n!}{j! (n-j)!} \\ &= \binom{2n}{n} \sum_{i=0}^n \left(\binom{n}{i} \right)^2 \\ &= \binom{2n}{n} \cdot \binom{2n}{n} = \left(\binom{2n}{n} \right)^2 \\ & P_{(0,0)(0,0)}^{(2n)} = \left(\binom{2n}{n} \right)^2 \cdot \frac{1}{16^n} \end{aligned}$$

Using Stirling's approximation:

$$\begin{aligned} \binom{2n}{n} &\sim \frac{4^n}{\sqrt{\pi n}} \Rightarrow \left(\binom{2n}{n} \right)^2 \sim \frac{16^n}{\pi n} \\ &\Rightarrow P_{(0,0)(0,0)}^{(2n)} \sim \frac{1}{n} \end{aligned}$$

So,

$$\sum_{n \in \mathbb{N}} P_{(0,0)(0,0)}^{(2n)} \sim \sum_{n=0}^{\infty} \frac{1}{n} \text{ diverges.}$$

Therefore, the chain is recurrent.

2.3 $d \geq 3$

Now the random walk can move to all the $2d$ directions with equal probability $\frac{1}{2d}$.

So, we have $P_{(0,0),\dots,(0,0)}^{(2n)}$ = probability that each co-ordinate of the random walk returns to 0 at time $2n$ which is equivalent to $\prod_{i=1}^d P(1 - D \text{ random in each co-ordinate } i \text{ returns to 0 at time } 2n_i)$, where each $\sum_{i=1}^d n_i = n$.

Hence we have,

$$\begin{aligned} P_{(0,0),\dots,(0,0)}^{(2n)} &= \prod_{i=1}^d \binom{2n_i}{n_i} 2^{-2n_i} \\ &\approx \prod_{i=1}^d \frac{1}{\sqrt{\pi n_i}} \\ &\leq \left(\frac{1}{\sqrt{\pi \frac{n}{d}}} \right)^d \\ &= \mathcal{O}(n^{-\frac{d}{2}}) \end{aligned}$$

Now $\sum_{n \in \mathbb{N}} P_{(0,0),\dots,(0,0)}^{(2n)} = \sum_{n \in \mathbb{N}} \mathcal{O}(n^{-\frac{d}{2}}) < \infty$, for $d \geq 3$, as $\sum_{n \in \mathbb{N}} \frac{1}{n^k} < \infty, \forall k > 1$.

Therefore, the chain is transient.

3 Random Walks on Electrical Networks:

Now we already know simple symmetric random walk on $\mathbb{Z}^d$ is recurrent for $d = 1, 2$ and transient for $d \geq 3$. Now suppose we remove an edge from $\mathbb{Z}^d$ or add an multi-edge or loop to $\mathbb{Z}^d$. Will this behaviour of recurrence or transience change?

For $\mathbb{Z}^d$, the answer is **No**. But the solution is not trivial because for other graphs rather than lattices, the recurrence or transience is affected by removing or adding edges.

This is a complicated question and to answer this we use the electrical networks.

3.1 Relation Between Finite Networks and Reversible Markov Chain:

A network is an undirected connected graph G with vertex set V and edge set E with corresponding edge-weights $\{c(e)\}, \forall e \in E$, called **conductances** associated to the edges of G .

Define $r(e) = \frac{1}{c(e)}$; $\forall e \in E$ which are called **resistances** associated with the edges of G . Clearly, $c(x, y) = c(y, x)$ and $r(x, y) = r(y, x)$ as conductances and resistances are defined on unoriented edges.

A network will be denoted by the pair $(G; \{c(e)\})$. Vertices of G are often called nodes. For $x, y \in V$, we will write $x \sim y$ to indicate that $\{x, y\}$ belongs to E .

Claim: The study of reversible Markov chain is equivalent to the study of random walks on networks.

Proof: Consider the Markov chain on the nodes of network G with transition matrix $P(x, y) = \frac{c(x, y)}{c(x)}$, where $c(x) = \sum_{y|y \sim x} c(x, y)$. This Markov chain is reversible with respect to the probability π defined by $\pi(x) := \frac{c(x)}{c_G}$, where $c_G = \sum_{x \in V} c(x)$:

$$\pi(x)P(x, y) = \frac{c(x)}{c_G} \cdot \frac{c(x, y)}{c(x)} = \frac{c(y)}{c_G} \cdot \frac{c(y, x)}{c(y)} = \pi(y)P(y, x)$$

and observing $\sum_{x \in V} \pi(x)P(x, y) = \frac{\sum_{x|x \sim y} c(x, y)}{c_G} = \pi(y)$, we say π is stationary.

This shows that every weighted random walk on a network is a reversible Markov chain. Now we shall show the reverse is also true, i.e; every reversible Markov chain is a weighted random walk on a network.

Suppose P is a transition matrix on a finite set χ which is reversible with respect to the probability π . Define a graph with vertex set χ by declaring $\{x, y\}$ an edge if $P(x, y) > 0$. This is a proper definition, since reversibility implies that $P(x, y) > 0$ exactly when $P(y, x) > 0$. Next, define conductances on edges by $c(x, y) = \pi(x)P(x, y)$. This is symmetric by reversibility. With this choice of weights, we have $c(x) = \pi(x)$, and thus the transition matrix associated with this network is just P .

Hence our claim is proved.

Note that, if the graph has no loops then we have $c_G = 2 \sum_{e \in E} c(e)$ and also observe that simple random walk is the special case when all edge weights are same.

3.2 Harmonic Functions:

To define harmonic functions we need to assume the Markov chain is **irreducible**. We call a function $F : \chi \rightarrow \mathbf{R}$ harmonic for P at a vertex x if $F(x) = \sum_{y \in \chi} P(x, y)F(y)$, i.e; observe that for simple random walk $F(x)$ is the average of the values of F at neighbouring vertices.

3.2.1 Existence, Uniqueness, Maximality Principle:

Theorem 3.1. *Let (X_t) be a Markov chain with irreducible transition matrix P , let $B \subseteq \chi$, and let $F_B : B \rightarrow \mathbb{R}$ be a function defined on B . Define $\tau_B = \min\{t \geq 0 | X_t \in B\}$. The function $F : \chi \rightarrow \mathbb{R}$ defined by $F(x) := \mathbb{E}_x F_B(X_{\tau_B})$ is the unique extension F of F_B to χ such that $F(x) = F_B(x)$ for all $x \in B$, and F is harmonic for P at all $x \in \chi \setminus B$.*

Proof. We first show that $F(x) = \mathbb{E}_x F_B(X_{\tau_B})$ is a harmonic extension of F_B . Clearly $F(x) = F_B(x)$ for all $x \in B$. Suppose that $x \in \chi \setminus B$. Then $F(x) = \mathbb{E}_x F(X_{\tau_B}) = \sum_{y \in \chi} P(x, y) \mathbb{E}_x[F(X_{\tau_B}) | X_1 = y]$. Observe that $x \in \chi \setminus B$ implies that $\tau_B \geq 1$. By the Markov property, it follows that $\mathbb{E}_x[F(X_{\tau_B}) | X_1 = y] = \mathbb{E}_y F(X_{\tau_B}) = F(y)$. Hence $F(x) = \mathbb{E}_x F(X_{\tau_B}) = \sum_{y \in \chi} P(x, y)F(y)$, i.e; F is harmonic.

We now show uniqueness. Suppose $g : \chi \rightarrow \mathbb{R}$ is a function which is harmonic on $\chi \setminus B$ and satisfies $g(x) = 0$ for all $x \in B$. We first show that $g \leq 0$. Define $A := \{u \in \chi : g(u) = \max_X g\}$. Fix $x \in A$. If $x \in B$, then $g \leq 0$ on X , so we

may assume that $x \notin B$. Suppose that $P(x, y) > 0$. Harmonicity of g on $\chi \setminus B$ implies that $g(x) = \sum_{z \in X} g(z)P(x, z) = g(y)P(x, y) + \sum_{z \in \chi \setminus \{y\}} g(z)P(x, z)$. If $g(y) < g(x)$, this would yield a contradiction, so we infer that $y \in A$.

By irreducibility, there exists a sequence of states $y_0, y_1, \dots, y_r$ such that $y_0 = x$, $y_r \in B$, each $y_i \notin B$ for $i < r$, and $P(y_{i-1}, y_i) > 0$ for $i = 1, 2, \dots, r$. Therefore, each $y_i \in A$; in particular, $y_r \in A$. Since $g(y_r) = 0$, it follows that $\max_X g = 0$. Applying this argument to $-g$ shows that $\min_X g \geq 0$, whence $g \equiv 0$ on X . Now, if F and $\tilde{F}$ are both harmonic on $\chi \setminus B$ and agree on B , then the difference $F - \tilde{F}$ is harmonic on $\chi \setminus B$ and vanishes on B . Therefore, $F(x) - \tilde{F}(x) = 0$ for all $x \in \chi$. $\square$

An example for simple random walk is shown below in figure-1, where the function was specified to be 1 at two vertices, 0.5 at another and 0 at a fourth vertex; the function is harmonic elsewhere.

Note: Harmonic functions are closed under linear combinations, hence these harmonic functions form an vector-space. Also note that requiring h to be harmonic on $\chi \setminus B$ yields a system of $(|\chi| - |B|)$ linear equations in the $(|\chi| - |B|)$ unknowns $\{F(x)\}_{x \in \chi \setminus B}$. For such a system of equations, existence of a solution implies uniqueness.

3.2.2 Examples:

The Heat function, Voltage function, the function representing the probability of ruin or win in a Gambler's ruin problem, the function representing Escape Probability all are some real-world examples of **harmonic** functions.

3.2.3 Matrix Representation For Complicated Cases:

Sometimes, we may face situations where the harmonic functions are not computable in hand by using the equation $F(x) = \sum_{y \in \chi} P(x, y)F(y)$. In those cases we solve it by using it's matrix representation $PF = F \Rightarrow (P - I)F = 0 \Rightarrow QF = 0$, where P is the transition matrix which is known, hence $Q = (P - I)$ matrix is known also and we evaluate F vector using the matrix representation through finding null-space of Q and then use the known **boundary values** to check which of those vectors belonging to the null-space represents the **unique harmonic function**. We can compute the null-space through R-coding.

```
1 install.packages("MASS")
2 library(MASS)
```

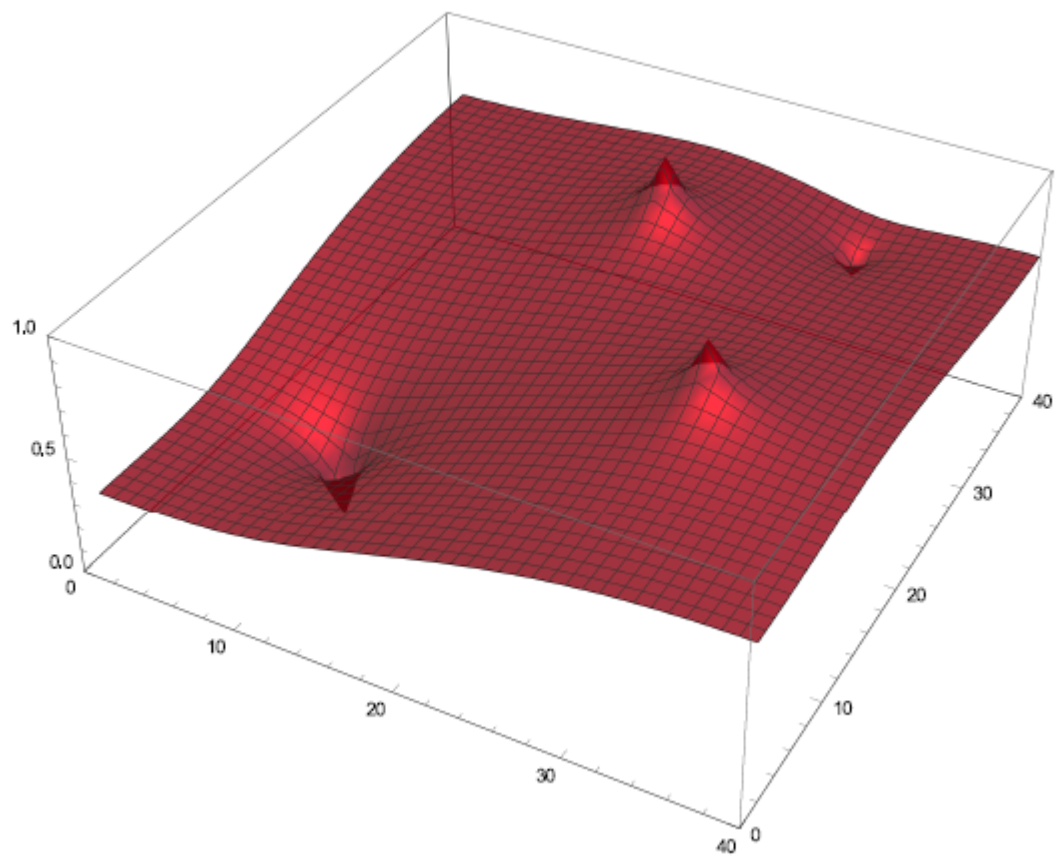


Figure 2: A harmonic function on a (40×40) square grid with 4 specified values where it is not harmonic

```

3 n<-
4 data<- c()                                #stores (n*n) values of transition matrix
   row by row
5 P<- matrix( data, nrow = n, byrow = TRUE)
6 Q<- (diag(n) - P)
7 null_space <- NULL(Q)
8 null_space
9
10 ###Now use boundary values to check which of the vectors
11 belonging to Nullspace is the unique harmonic function.###
12
13 boundary<- c()                            #a (2*1) vector

```

```
14 for(i in 1:n){  
15   if(null_space[i,1]=boundary[1] && null_space[i,n]=boundary[2]){  
16     print(null_space[i,])  
17   }  
18 }
```


3.3 Voltage and Current Flow on finite network:

Consider a network $(G; \{c(e)\})$. We distinguish two nodes, a and z , which are called the source and the sink of the network. A function W which is harmonic on $V \setminus \{a, z\}$ will be called a voltage. By the proposition on the harmonic functions we can see, a voltage is completely determined by its boundary values $W(a)$ and $W(z)$.

An oriented edge $\vec{e} = x\vec{y}$ is an ordered pair of nodes (x, y) . A flow ϕ is a function on **oriented edges which is antisymmetric**, meaning that $\phi(x\vec{y}) = -\phi(y\vec{x})$. For a flow ϕ , define the **divergence** of ϕ at x by $\text{div } \phi(x) := \sum_{y: y \sim x} \phi(x\vec{y})$. We note that for any flow ϕ we have $\sum_{x \in V} \text{div } \phi(x) = \sum_{x \in V} \sum_{y: y \sim x} \phi(x\vec{y}) = \sum_{\{x, y\} \in E} [\phi(x\vec{y}) + \phi(y\vec{x})] = 0$.

A **flow from a to z** is a flow ϕ satisfying: $\phi(x) = 0$ at all $x \notin \{a, z\}$ and $\text{div } \phi(a) \geq 0$. Hence observe that $\text{div } \phi(a) = -\text{div } \phi(z)$.

Strength of a flow ϕ from a to z is defined as $\|\phi\| := \text{div } \phi(a)$. A unit flow from a to z is a flow from a to z with strength 1.

Current Flow: Given a voltage W on the network, the current flow I associated with W is defined on oriented edges by $I(x\vec{y}) = \frac{W(x) - W(y)}{r(x\vec{y})} = c(x\vec{y})[W(x) - W(y)]$. The Current Flow is a flow as it is antisymmetric from definition and satisfies $\sum_{y: y \sim x} I(x\vec{y}) = \sum_{y: y \sim x} c(x, y)[W(x) - W(y)] = c(x)W(x) - c(x) \sum_{y: y \sim x} W(y)P(x, y) = 0, \forall x \notin \{a, z\}$.

The current flow satisfies several properties: **Ohm's Law:** If $x \sim y$, the current $i_{x,y}$ from x to y satisfies $W(x) - W(y) = I(x\vec{y})r(x\vec{y})$.

Kirchhoff's Node Law: The current is a flow between a and z .

Kirchhoff's Cycle Law: For finite network if $x_1 \sim x_2 \sim \dots \sim x_n \sim x_{n+1} = x_1$ is a cycle, then $\sum_{k=1}^n I(x_k \vec{x}_{k+1})r(x_k \vec{x}_{k+1}) = 0$.

Theorem 3.2. *If ϕ is a flow from a to z satisfying the cycle law $\sum_{i=1}^m r(\vec{e}_i)\phi(\vec{e}_i) = 0$ for any cycle $\vec{e}_1, \dots, \vec{e}_m$, and if $\|\phi\| = \|I\|$, then $\phi = I$.*

Proof. Consider the function $f = \phi - I$, observe that f satisfies the node law at all nodes and the cycle law. Suppose by contrary $f(\vec{e}_1) > 0$ for some oriented edge $\vec{e}_1$. By the node law, $\vec{e}_1$ must lead to some oriented edge $\vec{e}_2$ with $f(\vec{e}_2) > 0$. Iterating this process generates a sequence of oriented edges on which f is

strictly positive. Since the underlying network is finite, this sequence must eventually revisit a node. The resulting cycle violates the cycle law. Hence contradiction. So $f \equiv 0$, i.e; $\phi = I$. $\square$

3.4 Effective Resistance, Effective Conductance:

3.4.1 Definitions:

Given a network, the ratio $\frac{W(a)-W(z)}{\|I\|}$, where I is the current flow corresponding to the voltage W , is independent of the voltage W applied to the network. Define the effective resistance between vertices a and z by $R(a \leftrightarrow z) := \frac{W(a)-W(z)}{\|I\|}$. (Imagine replacing our entire network by a single edge joining a to z with resistance $R(a \leftrightarrow z)$. If we now apply the same voltage to a and z in both networks, then the amount of current owing from a to z in the single-edge network is the same as in the original. Hence $R(a \leftrightarrow z)$ is called **effective resistance**.) In parallel with our earlier definitions, we also define the **effective conductance** as $C(a \leftrightarrow z) = \frac{1}{R(a \leftrightarrow z)}$.

3.4.2 Network Reduction:

Series Law: Suppose $v \in V \setminus \{a, z\}$ be a node of degree 2 with neighbors u_1, u_2 and we replace the edges (u_1, v) and (v, u_2) by a single edge (u_1, u_2) having resistance $r(u_1 v) + r(v u_2)$. All potentials and currents in $G \setminus \{v\}$ and the current that flows from u_1 to u_2 equals $I(u_1 v) = I(v u_2)$.

Parallel Law: If two edges e_1 and e_2 that both join vertices $u_1, u_2 \in V$ are replaced by a single edge e joining u_1, u_2 with conductance $c_e := c_{e_1} + c_{e_2}$, then all voltages and currents in $G \setminus \{e_1, e_2\}$ are unchanged, and the current on that single edge equals $I(e_1) + I(e_2)$.

Gluing: We define the network operation of gluing vertices v and w by identifying v and w and keeping all existing edges, i.e; any edges between v and w become loops. If the voltages at v and w are the same and v and w are glued, then because current never flows between vertices with the same voltage, voltages and currents are unchanged.

Star-Triangle Law: The configurations in the Figure-2 are equivalent when $\forall i \in \{1, 2, 3\}$,

$$c(w, u_i)c(u_{i-1}, u_{i+1}) = \varphi, \text{ where indices are taken modulo 3, and } \varphi := \frac{\prod_i c(w, u_i)}{\sum_i c(w, u_i)} =$$

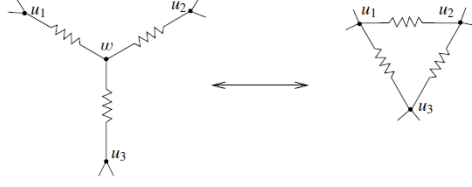


Figure 3: Star-Triangle configuration

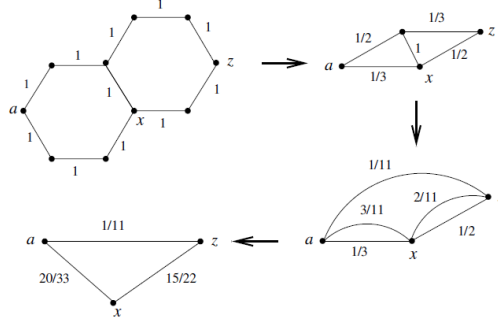


Figure 4: Figure-3

$$\frac{\sum_i r(u_{i-1}, u_{i+1})}{\prod_i r(u_{i-1}, u_{i+1})}.$$

Example: For example observe that in Figure-3 we reduced the network by parallel law in first step, by Star-triangular law in second step and then again applying parallel law in third step.

Note: When a and z are two vertices in a tree Γ with unit resistance on each edge, then $R(a \leftrightarrow z)$ is equal to the length of the unique path joining a and z . (For any vertex x not along the path joining a and z , there is a unique path from x to a . Let x_0 be the vertex at which the x - a path first hits the a - z path. Then $W(x) = W(x_0)$.)

3.4.3 Examples:

Example-1: For a tree Γ with root ρ , let Γ_n be the set of vertices at distance n from ρ . Consider the case of a spherically symmetric tree, in which all vertices of Γ_n have the same degree for all $n \geq 0$. Suppose that all edges at the same distance from the root have the same resistance, that is, $r(e) = r_i$ if the vertex of e furthest from the root is at distance i to the root, $i \geq 1$. Glue all the vertices in each level; this will not affect effective resistances, so we infer that $R(\rho \leftrightarrow \Gamma_M) = \sum_{i=1}^M \frac{r_i}{|\Gamma_i|}$, and

$$P_\rho\{\tau_{\Gamma_M} < \tau_\rho^+\} = \frac{r_1/|\Gamma_1|}{\sum_{i=1}^M r_i/|\Gamma_i|}.$$

Therefore, $\lim_{M \rightarrow \infty} P_\rho\{\tau_{\Gamma_M} < \tau_\rho^+\} > 0$ if and only if $\sum_{i=1}^\infty \frac{r_i}{|\Gamma_i|} < \infty$.

Example-2 (Biased nearest-neighbor random walk): Fix $\alpha > 0$ with $\alpha \neq 1$ and consider the path with vertices $\{0, 1, \dots, n\}$ and conductances $c(k-1, k) = \alpha^k$ for $k = 1, \dots, n$. Then for all interior vertices $0 < k < n$, we have $P(k, k+1) = \frac{\alpha}{1+\alpha}$, $P(k, k-1) = \frac{1}{1+\alpha}$. If $p = \frac{\alpha}{1+\alpha}$, then this is the walk that, when at an interior vertex, moves up with probability p and down with probability $1-p$. Using the Series Law, we can replace the k edges to the left of vertex k by a single edge of resistance $r_1 := \sum_{j=1}^k \alpha^{-j} = \frac{1-\alpha^{-k}}{\alpha-1}$. Likewise, we can replace the $(n-k)$ edges to the right of k by a single edge of resistance $r_2 := \sum_{j=k+1}^n \alpha^{-j} = \frac{\alpha^{-k}-\alpha^{-n}}{\alpha-1}$. The probability $P_k\{\tau_n < \tau_0\}$ is not changed by this modification, so we can calculate simply that $P_k\{\tau_n < \tau_0\} = \frac{r_2^{-1}}{r_1^{-1}+r_2^{-1}} = \frac{\alpha^{-k}-1}{\alpha^{-n}-1}$.

In particular, for the biased random walk which moves up with probability p , $P_k\{\tau_n < \tau_0\} = \frac{[(1-p)/p]^k - 1}{[(1-p)/p]^n - 1}$.

3.5 Energy of a Flow, Green's function, Escape Probability:

3.5.1 Definitions:

Define the energy of a flow ϕ by $E(\phi) := \sum_e [\phi(e)]^2 r(e)$.

The Green's function for a random walk stopped at a stopping time τ is defined by $G_\tau(a, x) := \mathbb{E}_a(\text{number of visits to } x \text{ before } \tau) = \mathbb{E}_a(\sum_{t=0}^{\infty} \mathbf{1}\{X_t = x, \tau > t\})$.

The escape probability $P_a(\tau_z < \tau_a^+)$ is the probability that a walker started at a hits z before returning to a .

3.5.2 Lemmas:

Theorem 3.3. *For any $a, z \in X$ with $a \neq z$, the escape probability is given by: $P_a(\tau_z < \tau_a^+) = \frac{1}{c(a)R(a \leftrightarrow z)} = \frac{C(a \leftrightarrow z)}{c(a)}$.*

Proof. Take $B = \{a, z\}$ and $h_B = \mathbb{1}_{\{z\}}$. Applying proposition-1 here, we get that $x \mapsto \mathbb{E}_x h_B(X_{\tau_B}) = \mathbb{P}_x(\tau_z < \tau_a)$ is the unique harmonic function on $X \setminus \{a, z\}$ with boundary values 0 at a and 1 at z . Since the function $x \mapsto \frac{W(a) - W(x)}{W(a) - W(z)}$ is also harmonic on $X \setminus \{a, z\}$ with the same boundary values, Proposition-1 implies that: $\mathbb{P}_x(\tau_z < \tau_a) = \frac{W(a) - W(x)}{W(a) - W(z)}$. Therefore, $P_a(\tau_z < \tau_a^+) = \sum_{x \in V} P(a, x) \mathbb{P}_x(\tau_z < \tau_a)$.

Using the transition probabilities and conductances, we can write: $P_a(\tau_z < \tau_a^+) = \sum_{x: x \sim a} \frac{c(a, x)}{c(a)} \cdot \frac{W(a) - W(x)}{W(a) - W(z)}$. By the definition of current flow I the numerator becomes: $\sum_{x: x \sim a} I(a \vec{x}) = \|I\|$, and thus: $P_a(\tau_z < \tau_a^+) = \frac{\|I\|}{c(a)[W(a) - W(z)]} = \frac{1}{c(a)R(a \leftrightarrow z)}$. $\square$

Theorem 3.4. *Green's function satisfies, $G_{\tau_z}(a; a) = c(a)R(a \leftrightarrow z)$.*

Proof. We know $G_{\tau_z}(a; x) = \mathbb{E}_a(\text{number of visits to } x \text{ before } \tau_z)$. In particular, $G_{\tau_z}(a; a)$ is the expected number of returns to a before hitting z , starting from a where the no. of visits to a before visiting z follows a Geometric distribution with parameter $P_a(\tau_z < \tau_a^+)$. Hence $G_{\tau_z}(a; a) = \frac{1}{P_a(\tau_z < \tau_a^+)} = c(a)R(a \leftrightarrow z)$. $\square$

3.5.3 Transience Holds $\Leftrightarrow$ Finite Resistance:

We know $P_a(\tau_z < \tau_a^+) = \frac{1}{c(a)R(a \leftrightarrow z)}$.

A random walk on network is transient $\Leftrightarrow P_a(\tau_a^+ < \infty) < 1 \Leftrightarrow (1 - P_a(\tau_a^+ > \infty)) < 1 \Leftrightarrow P_a(\tau_a^+ > \infty) > 0$.

For an infinite network G , let $\langle G_n \rangle$ be any sequence of finite sub-graphs of G such that $G_n \subset G_{n+1}$ and $G = \cup_n G_n$. Each edge in G_n is an edge in G , so we simply give it the same conductance it has in G . We also assume that G_n is the graph induced in G by $V(G_n)$. Let Z_n be the set of vertices in $G \setminus G_n$. Let G_n^W be the graph obtained from G by identifying Z_n to a single vertex z_n and then removing loops (but keeping multiple edges). Given a network G , consider a random walk on it, if we stop it the 1st time it reaches Z_n , then we obtain a random walk on G_n^W until it reaches z_n . Now for every $a \in G$, the events $[a \rightarrow Z_n]$ is decreasing in n as $Z_{n+1} \subset Z_n$ and hence $P(a \rightarrow Z_n) = P_a(\tau_{Z_n} < \tau_a^+)$ is decreasing in n . Now as $P_a(\tau_{Z_n} < \tau_a^+)$ is a probability, it is bounded in $[0, 1]$, hence bounded below by 0, so by monotone convergence theorem $P_a(\tau_{Z_n} < \tau_a^+)$ converges as $n \rightarrow \infty$ and $\lim_{n \rightarrow \infty} P_a(\tau_{Z_n} < \tau_a^+) = P_a(\tau_a^+ > \infty) = \frac{1}{c(a)R(a \leftrightarrow \infty)}$. Hence the random walk is transient $\Leftrightarrow R(a \leftrightarrow \infty) < \infty$ (i.e; finite resistance).

3.6 Thompson's Principle and Rayleigh's Monotonicity:

3.6.1 Thompson's Principle:

Theorem 3.5. *For any finite connected graph, the effective resistance $R(a \leftrightarrow z)$ is given by: $R(a \leftrightarrow z) = \inf \{E(\phi) : \phi \text{ is a unit flow from } a \text{ to } z\}$. (The unique minimizer in the infimum above is the unit current flow.)*

Proof. Fixing some unit flow ϕ_0 from a to z , consider the set $K = \{\psi \text{ unit flows from } a \text{ to } z : E(\psi) \leq E(\psi_0)\}$. This is a compact subset of $\mathbf{R}^{|E|}$ as K is bounded and its complement is open in $\mathbf{R}$. Therefore, there exists a unit flow ϕ from a to z minimizing $E(\phi)$ subject to $\|\phi\| = 1$. By Proposition-2, to prove that the unit current flow is the unique minimizer, it suffices to verify that any unit flow ϕ of minimal energy satisfies the cycle law.

Let the edges $\vec{e}_1, \dots, \vec{e}_n$ form a cycle. Define $\varphi(\vec{e}_i) = 1$ for all $1 \leq i \leq n$, and set φ equal to zero on all other edges. Note that φ satisfies the node law, so it is a flow. However, $\sum \varphi(\vec{e}_i) = n \neq 0$. For any $\epsilon \in \mathbb{R}$, the minimality of $E(\phi)$ implies that: $0 \leq E(\phi + \epsilon\varphi) - E(\phi) = \sum_{i=1}^n [(\phi(\vec{e}_i) + \epsilon)^2 - \phi(\vec{e}_i)^2] r(\vec{e}_i)$.

Expanding the terms gives: $0 \leq 2\epsilon \sum_{i=1}^n r(\vec{e}_i)\phi(\vec{e}_i) + O(\epsilon^2)$.

Dividing through by $\epsilon > 0$ and letting $\epsilon \downarrow 0$ gives: $0 \leq 2 \sum_{i=1}^n r(\vec{e}_i)\phi(\vec{e}_i)$. Similarly, dividing through by $\epsilon < 0$ and letting $\epsilon \uparrow 0$ yields: $0 \geq 2 \sum_{i=1}^n r(\vec{e}_i)\phi(\vec{e}_i)$. Thus: $\sum_{i=1}^n r(\vec{e}_i)\phi(\vec{e}_i) = 0$, verifying that ϕ satisfies the cycle law.

Finally, we show that the unit current flow I satisfies $E(I) = R(a \leftrightarrow z)$.

By definition: $E(I) = \sum_e r(e)I(e)^2 = \frac{1}{2} \sum_x \sum_y r(x, y) \left(\frac{W(x) - W(y)}{r(x, y)} \right)^2 = E(I) = \frac{1}{2} \sum_x \sum_y c(x, y) [W(x) - W(y)]^2$.

By anti-symmetry of I , we rewrite: $\frac{1}{2} \sum_x \sum_y [W(x) - W(y)] I(\vec{xy}) = \sum_x W(x) \sum_y I(\vec{xy})$.

Using the node law, $\sum_y I(\vec{xy}) = 0$ for all $x \notin \{a, z\}$, while $\sum_y I(\vec{ay}) = \|I\| = -\sum_y I(\vec{zy})$. Therefore, the right-hand side reduces to: $\|I\|(W(a) - W(z)) = \frac{W(a) - W(z)}{\|I\|}$.

Since $\|I\| = 1$, we conclude: $E(I) = R(a \leftrightarrow z)$.

If $r = r(e)$ are assignments of resistances to the edges of a graph G , write $R(a \leftrightarrow z; r)$ to denote effective resistance. Now we shall prove Rayleigh's monotonicity law and some corollaries from which we decide how escape probability is affected if we add an edge that is not incident to a . $\square$

3.6.2 Rayleigh's Monotonicity Law:

Theorem 3.6. *If $\{r(e)\}$ and $\{r'(e)\}$ are two assignments of resistances to the edges of the same finite network G such that $r(e) \leq r'(e)$ for all e , then: $R(a \leftrightarrow z; r) \leq R(a \leftrightarrow z; r')$.*

Proof. Note that $\inf_{\phi} \sum_e r(e) \phi(e)^2 \leq \inf_{\phi} \sum_e r'(e) \phi(e)^2$, where the infimum are taken over all unit flows ϕ . This inequality holds because for each edge e , $r(e) \leq r'(e)$. Now by Thompson's principle the rest of the proof follows. $\square$

Corollary 3.7. *Adding an edge in a finite network does not increase the effective resistance $R(a \leftrightarrow z)$. If the added edge is not incident to a , then the addition does not decrease the escape probability $P_a[\tau_z < \tau_a^+] = [c(a)R(a \leftrightarrow z)]^{-1}$.*

Proof. Before we add an edge to a network, we can think of it as existing already with $c = 0$ or $r = \infty$, which corresponds to no flow along that edge (i.e., infinite resistance). Adding the edge reduces its resistance to a finite value. Hence by Rayleigh's monotonicity law, $R(a \leftrightarrow z)$ doesn't increase after adding an edge.

Now we know, the escape probability is can be written as $P_a[\tau_z < \tau_a^+] = [c(a)R(a \leftrightarrow z)]^{-1} = \frac{C(a \leftrightarrow z)}{c(a)}$. Now if the added edge isn't incident to a , we regard it as changing a conductance from 0 to a positive no. and hence the escape probability can't decrease. $\square$

Corollary 3.8. *The operation of gluing vertices cannot increase effective resistance.*

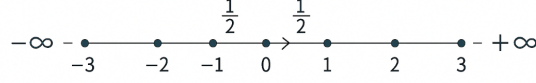


Figure 5: $\mathbb{Z}$

Proof. When we glue vertices together, we take an infimum in Thomson's Principle over a larger class of flows and hence effective resistance can't increase. $\square$

3.6.3 Computing Effective Resistance for $\mathbb{Z}, \mathbb{Z}^2$ Lattices:

- $\mathbb{Z}$:

Consider simple symmetric random walk on $\mathbb{Z}$.

Hence each edge conductance is 1 for all edges and each vertex being adjacent to 2 edges each vertex-conductance becomes 2. (See figure-5)

Now for any integer point a , $P_a(\tau_{a+1} < \tau_a^+) = \frac{1}{2} = P_a(\tau_{a-1} < \tau_a^+)$.

So, $R(a \longleftrightarrow a+1) = \frac{1}{2 \cdot \frac{1}{2}} = 1$.

Hence applying series law, $R(a \longleftrightarrow \infty) = \sum_{i=a}^{\infty} R(a \longleftrightarrow a+1) = \infty$.

Hence walk is recurrent.

- $\mathbb{Z}^2$:

Consider simple symmetric random walk on $\mathbb{Z}^2$.

Hence, each edge conductance is 1 for all edges and being adjacent to 4 edges, each vertex-conductance becomes 4. (See figure-6)

In figure-6, consider the bolded squares as vertices and consider centre-most point as 0, just it's outer bolded square as point 1 making all edge-conductances at distance 1 equal to 0 and keep doing so on. So, 0 is connected to 1 with 4

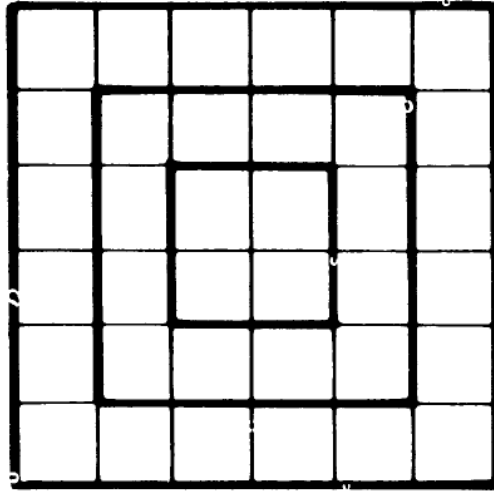


Figure 6: Sorting These Squares As A Point Making Conductances of The Edges Zero

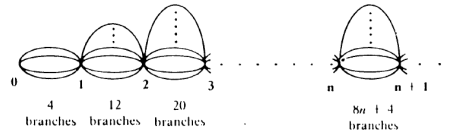


Figure 7: After Sorting We Get The Above Structure

edges, 1 is connected to 2 with 12 edges and similarly $(k - 1)$ is connected to k with $(8k - 4)$ edges for any k .

Now make all the edge-conductances of all the bolded squares 0, then we get figure-7 after doing this.

Now applying parallel resistance law, we get the resistances between any two vertices $(k - 1)$ and k as $\frac{1}{(8k-4)}$. (See figure-8)

Now applying series law, we get $R(0 \longleftrightarrow \infty) = \sum_{k=1}^{\infty} \frac{1}{(8k-4)} = \infty$.

Now by Rayleigh's monotonicity principle, effective resistance of this network is smaller than $\mathbb{Z}^2$, hence effective resistance is infinite for $\mathbb{Z}^2$ as well.

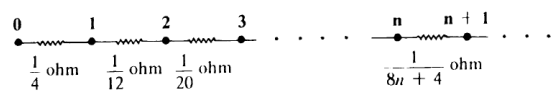


Figure 8: After Applying Parallel Law of Resistance

So, simple symmetric random walk on $\mathbb{Z}^2$ is recurrent.

3.7 Nash-William's inequality and Cutset:

3.7.1 Nash-William's Inequality:

Call $\Pi \subseteq E$ an edge-cutset separating a to z if every path from a to z includes some edge in Π .

Theorem 3.9. *If ϕ is a flow from a to z , and Π is an edge-cutset separating a from z , then $\|\phi\| \leq \sum_{e \in \Pi} |\phi(e)|$.*

Proof. Define the set $S = \{x \in V : a \text{ and } x \text{ are connected in } G \setminus \Pi\}$. By the definition of a flow, the divergence of ϕ at each vertex $x \neq a, z$ satisfies $\text{div } \phi(x) = 0$. Thus, summing over all vertices in S , $\sum_{x \in S} \sum_y \phi(\vec{xy}) = 0$, where $\phi(\vec{xy})$ is the flow along the oriented edge $\vec{xy}$. The total flow leaving S through the edges in Π is given by $\sum_{x \in S} \sum_{y \notin S} \phi(\vec{xy})$, which corresponds to the net flow crossing the edge-cutset Π . By the definition of Π , every path from a to z must cross some edge in Π , ensuring that all flow from a to z is accounted for in this sum. Therefore, $\|\phi\| = \sum_{x \in S} \sum_{y \notin S} \phi(\vec{xy})$, and by the triangle inequality, $\|\phi\| \leq \sum_{e \in \Pi} |\phi(e)|$. $\square$

Theorem 3.10. Nash-Williams Inequality: *If $\{\Pi_k\}$ are disjoint edge-cutsets separating nodes a and z , then $R(a \leftrightarrow z) \geq \left(\sum_k (\sum_{e \in \Pi_k} c(e))^{-1}\right)^{-1}$.*

Proof. Let ϕ be a unit flow from a to z . For any cutset Π_k , applying the Cauchy-Schwartz inequality gives $(\sum_{e \in \Pi_k} c(e)) (\sum_{e \in \Pi_k} r(e) \phi(e)^2) \geq (\sum_{e \in \Pi_k} |\phi(e)|)^2$, since $c(e) = 1/r(e)$. we can rewrite this as $\sum_{e \in \Pi_k} r(e) \phi(e)^2 \geq \frac{(\sum_{e \in \Pi_k} |\phi(e)|)^2}{\sum_{e \in \Pi_k} c(e)}$. By Proposition-5, the total flow through Π_k satisfies $\sum_{e \in \Pi_k} |\phi(e)| \geq \|\phi\| = 1$, where $\|\phi\| = 1$ since ϕ is a unit flow. Substituting this lower bound, we get $\sum_{e \in \Pi_k} r(e) \phi(e)^2 \geq \frac{1}{\sum_{e \in \Pi_k} c(e)}$. Summing over all disjoint cutsets $\{\Pi_k\}$, we have $\sum_{e \in E} r(e) \phi(e)^2 \geq \sum_k \frac{1}{\sum_{e \in \Pi_k} c(e)}$. By Thomson's Principle, the effective resistance $R(a \leftrightarrow z)$ is the minimum energy over all unit flows, so $R(a \leftrightarrow z) = \inf_{\phi} \sum_{e \in E} r(e) \phi(e)^2$. Thus, $R(a \leftrightarrow z) \geq \sum_k \frac{1}{\sum_{e \in \Pi_k} c(e)} = \left(\sum_k (\sum_{e \in \Pi_k} c(e))^{-1}\right)^{-1}$.

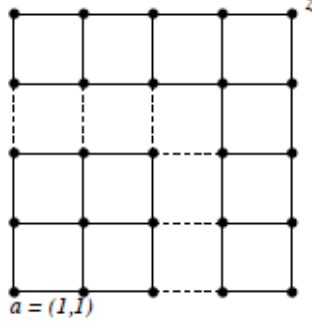


Figure 9: Graph B_5 , where dashed lines form cutset Π_3

□

3.7.2 Escape Probability on a Square:

We now use the inequalities we have developed to bound effective resistance in a non-trivial example. Let B_n be the $n \times n$ two-dimensional grid graph: the vertices are pairs of integers (z, w) such that $1 \leq z, w \leq n$, while the edges are pairs of points at unit (Euclidean) distance.

Theorem 3.11. *Let $a = (1, 1)$ be the lower left-hand corner of B_n , and let $z = (n, n)$ be the upper right-hand corner of B_n . Suppose each edge of B_n has unit conductance. The effective resistance $R(a \leftrightarrow z)$ satisfies $\frac{\log n}{2} \leq R(a \leftrightarrow z) \leq 2 \log n$.*

Proof. First we prove the lower bound.

Let Π_k be the edge set $\Pi_k = \{\{v, w\} \in E(B_n) : \|v\|_1 = k, \|w\|_1 = k+1\}$, where $\|(v_1, v_2)\|_1 = \max\{v_1, v_2\}$. (See Figure-4) Since every path from a to z must use an edge in Π_k , the set Π_k is a cutset. Since each edge has unit conductance, $\sum_{e \in \Pi_k} c(e)$ equals the number of edges in Π_k , namely $2k$. By Proposition-6 and an inequality of logarithm property, $R(a \leftrightarrow z) \geq \sum_{k=1}^{n-1} \frac{1}{2k} \geq \frac{\log n}{2}$.

Now we prove for the upper bound.

By **Thomson's Principle** the effective resistance is the minimal possible energy of a unit flow from a to z . To get an upper bound on resistance, we build a unit flow on the square.

Consider Polya's urn process. The sequence of ordered pairs listing the numbers of black and white balls is a Markov chain with state space $\{1, 2, \dots\}^2$. Run this process on the square—note that it necessarily starts at vertex $a = (1, 1)$ and stop when you reach the main diagonal $x + y = n + 1$. Direct all edges of the square from bottom left to top right and give each edge e on the bottom left half of the square the flow $f(e) = \mathbb{P}(\text{the process went through } e)$. To finish the construction, give the upper right half of the square the symmetrical flow values. From Polya's urn process, it follows that for any $k \geq 0$, the Polya's urn process is equally likely to pass through each of the $k + 1$ pairs (i, j) for which $i + j = k + 2$. Consequently, when (i, j) is a vertex in the square for which $i + j = k + 2$, the sum of the flows on its incoming edges is $\frac{1}{k+1}$. Thus, the energy of the flow f can be bounded by $E(f) \leq \sum_{k=1}^{n-1} 2 \left(\frac{1}{k+1}\right)^2 (k+1) \leq 2 \log n$. Hence our proof is complete. □

3.7.3 Polya's Theorem:

Theorem 3.12. *Simple Symmetric Random Walk is Transient for $d \geq 3$.*

Proof. Consider an urn with d balls each of a different colour. At each unit of time draw a ball at random and replace along-with an additional ball of the same colour. Let $B_t^{(j)}$ be the number of balls of colour j after t draws. So $B_0^{(j)} = 1, \forall j = 1, 2, \dots, d$.

Let $d \geq 2$ and $U_{-d+1} = 1$. Let $U_{-d+2}, U_{-d+3}, \dots, U_0, U_1, \dots$ be iid and uniform on $[0, 1]$. Let $V_1 \leq V_2 \leq \dots \leq V_d$ be the order statistics for $U_{-d+1}, \dots, U_0$, i.e; V_j is the j th smallest among these variables. Consider $V_0 = 0$ and define $A_t^{(j)} = |\{-d+1 \leq k \leq t | V_{j-1} < U_k \leq V_j\}|, \forall j = 1, 2, \dots, d$, observe $A_0^{(j)} = 1, \forall j = 1, 2, \dots, d$.

Claim: The distribution of $B_t^{(j)}$ and $A_t^{(j)}$ are same.

Proof of the claim: Notice that they begin with same distribution, so it is enough to show that, transition probabilities for $B_t^{(j)}$ are equal to those of $A_t^{(j)}$.

Conditioned on the relative positions of $(U_{-d+2}, \dots, U_t)$, the relative position of U_{t+1} is uniform on all $(t + d)$ possibilities. Hence the conditional probability

that U_{t+1} falls between V_{j-1} and V_j is proportional to the number of uniform random variables among $U_0, U_1, \dots, U_t$ which fall in this interval plus 1. Thus conditional probability that $A_t^{(j)}$ increases by 1 is equal to $\frac{A_t^{(j)}}{(t+d)}$ and by mechanism it is same as transition probability in $B_t^{(j)}$ to increase by 1. Thus our claim is proved. $\square$

Now $(A_t^{(1)}, A_t^{(2)}, \dots, A_t^{(d)})$ is uniform over $\{(x_1, x_2, \dots, x_d) | \sum_{i=1}^d x_i = (t+d), x_i \geq 1, \forall i = 1, 2, \dots, d\}$.

Construct a flow on the box $\{1, 2, \dots, n\}^d$ by defining for edges in the lower half of the box, $\theta(e) = \mathbb{P}(\text{Polya's } d\text{-coloured process goes through } e)$. We know that the process is equally likely to pass through each d -tuple $(x_1, x_2, \dots, x_d)$ with $\sum_{i=1}^d x_i = (k+d)$. Now there are $\binom{k+d-1}{d-1}$ such d -tuples. Hence each such edge has flow $(\binom{k+d-1}{d-1})^{-1}$. Now there exists c_1, c_2 (depended on d) such that $c_1 \leq \frac{\binom{k+d-1}{d-1}}{k^{d-1}} \leq c_2$. So energy $\varepsilon(\theta) \leq 2 \sum_{k=1}^{n-1} (\binom{k+d-1}{d-1})^{-2} \binom{k+d-1}{d-1} \leq c_3(d) \sum_{k=1}^{n-1} k^{-d+1} \leq c_4(d), \forall d \geq 3$. Hence $\forall a \in \mathbb{Z}^d, d \geq 3, \mathcal{R}(a \longleftrightarrow \infty) < \infty$ if and only if $\mathbb{P}_a(\tau_a^+ < \infty) < 1$, which implies transience. $\square$

4 Hitting Times and Cover Times:

Let (X_t) be a finite Markov chain with state-space χ . The cover-time variable τ_{cov} of (X_t) is the 1st time at which all the states have been visited, i.e; it is the minimal value such that for every state $y \in \chi$, $\exists t \leq \tau_{cov}$ with $X_t = y$. Define cover-time t_{cov} as $t_{cov} = \max_{x \in \chi} \mathbb{E}_x(\tau_{cov})$.

Use: For random walks on networks, cover time provides an estimate of the running time.

Example :Cover-time of cycle : Lovasz gives an elegant computation of the cover time t_{cov} of simple random walk on the n -cycle. This walk is simply the remainder modulo n of a random walk on $\mathbf{Z}$. The walk on the remainders has covered all n states exactly when the walk on $\mathbf{Z}$ has first visited n distinct states. Let c_n be the expected value of the time when a simple random walk on $\mathbf{Z}$ has first visited n distinct states and consider a walk which has just reached it's $(n - 1)$ st new state. The visited states form a subsegment of the number line and the walk must be at one end of that segment. Reaching the n th new state is now a gambler's ruin situation: the walker's position corresponds to a fortune of 1 (or $(n - 1)$) and we are waiting for her to reach either 0 or n . Either way, the expected time is $(n - 1)(1) = (n - 1)$. Hence $c_n = c_{n-1} + (n - 1) \Rightarrow c_n = \frac{n(n-1)}{2}$.

4.1 Giving Upper and Lower Bounds to Cover Times Using Hitting Times:

Since any walk started at x must have visited y by the time all states are covered, we have $t_{hit} = \max_{x,y \in \chi} \mathbb{E}_x(\tau_y) \leq \max_{x \in \chi} \mathbb{E}_x(\tau_{cov}) = t_{cov}$

Now to give upper bounds on cover times we basically use the concept that a walk covering all the states can visit them in many different orders and give a tighter lower bound using this concept on subsets of the state-space. This method is known as **Matthew's Method**.

Theorem 4.1. *Let (X_t) be an irreducible finite Markov chain on $n > 1$ states. Then $t_{cov} \leq t_{hit}(1 + \frac{1}{2} + \dots + \frac{1}{(n-1)})$.*

Proof. Without loss of generality, we may assume that our state space is $\{1, \dots, n\}$ and our starting state is n . Let σ be a uniform random permutation of $\{1, 2, \dots, n-1\}$, chosen independently of the chain. We will look for states in

the order σ . Let T_k be the first time that the states $\sigma(1), \dots, \sigma(k)$ have all been visited, and let $L_k = X_{T_k}$ be the last state among $\sigma(1), \dots, \sigma(k)$ to be visited. For any $1 \leq s \leq n-1$, we have $\mathbb{E}_n(T_1 \mid \sigma(1) = s) = \mathbb{E}_n(\tau_s) \leq t_{\text{hit}}$. Averaging over s shows that $\mathbb{E}_x(T_1) \leq t_{\text{hit}}$. For any choice of distinct $1 \leq r \neq s \leq n-1$, we have $\mathbb{E}_n(T_k - T_{k-1} \mid L_{k-1} = r, \sigma(k) = s = L_k) = \mathbb{E}_r(\tau_s) \leq t_{\text{hit}}$. Averaging over r and s yields $\mathbb{E}_n(T_k - T_{k-1} \mid L_k = \sigma(k)) \leq t_{\text{hit}}$. Observe that, for any set S of k elements, we have $\mathbb{P}_n\{L_k = \sigma(k) \mid \{\sigma(1), \dots, \sigma(k)\} = S, \{X_t\}_t = \frac{1}{k}\}$. Now $\mathbb{E}_n(T_k - T_{k-1} \mid L_k \neq \sigma(k)) = 0$. Hence $\mathbb{E}_n(T_k - T_{k-1}) = \mathbb{E}_n(T_k - T_{k-1} \mid L_k = \sigma(k))\mathbb{P}_n(L_k = \sigma(k)) \leq \mathbb{P}_n(L_k = \sigma(k)) \cdot t_{\text{hit}} = \frac{1}{k}t_{\text{hit}}$. Therefore, $t_{\text{cov}} = \mathbb{E}_n(T_{n-1}) \leq t_{\text{hit}} \sum_{k=1}^{n-1} \frac{1}{k}$. $\square$

Theorem 4.2. Let $A \subseteq \chi$. Set $t_{\min}^A = \min_{a,b \in A, a \neq b} \mathbb{E}_a(\tau_b)$. Then $t_{\text{cov}} \geq t_{\min}^A(1 + \frac{1}{2} + \dots + \frac{1}{|A|-1})$.

Proof. Fix an initial state $x \in A$. Let σ be an uniform random permutation of the elements of A chosen independently of the chain-trajectory.

Let T_k be the 1st time at which all of $\sigma(1), \sigma(2), \dots, \sigma(k)$ have been visited. Let $L_k = X_{T_k}$, now with probability $\frac{1}{|A|}$, we have $\sigma(1) = x$ and $T_1 = 0$. Otherwise, the walk must proceed from x to $\sigma(1)$. Thus $\mathbb{E}_x(T_1) = \sum_{s \in A} \mathbb{E}_x(T_1 \mid \sigma(1) = s)\mathbb{P}_x(\sigma(1) = s) = \frac{1}{|A|}(0 + \sum_{s \neq x} \mathbb{E}_x(T_1 \mid \sigma(1) = s)) \geq \frac{(|A|-1)}{|A|}t_{\min}^A$.

Now for $2 \leq k \leq |A|$ and $r, s \in A$, we have $\mathbb{E}_x(T_k - T_{k-1} \mid \sigma(k-1) = r, \sigma(k) = s = L_k) = \mathbb{E}_r(\tau_s) \geq t_{\min}^A$. Hence averaging over r, s , we get $\mathbb{E}_x(T_k - T_{k-1} \mid \sigma(k) = L_k) \geq t_{\min}^A$.

Now as $\mathbb{E}_x(T_k - T_{k-1} \mid \sigma(k) \neq L_k) = 0$, hence $\mathbb{E}_x(T_k - T_{k-1}) = \mathbb{E}_x(T_k - T_{k-1} \mid \sigma(k) = L_k)\mathbb{P}_x(L_k = \sigma(k)) \geq \frac{1}{k}t_{\min}^A$.

Hence, $t_{\text{cov}} \geq \mathbb{E}_x(\tau_{\text{cov}}) \geq \mathbb{E}_x(\tau_{\text{cov}}^A), \forall x \in A$. $\square$

4.2 Hitting and Cover Time for Different Graphs:

4.2.1 n-length path:

Hitting time: $t_{\text{hit}} = \max_{x,y} E_x(\tau_y) = \max_{x,y \in \{0,1,\dots,n\}} x(y-x)$, which is $\frac{n^2}{4}$ for n even and $\frac{n^2-1}{4}$ for n odd.

Cover time: Let t_n be the expected time it visits all the n states, i.e; cover time. Suppose it has visited 0 to $(n-1)$ th state in expected time $t_{(n-1)}$. Then from $(n-1)$ to reach n th state the expected time is say f_{n-1} where $f_k = \frac{(f_{k-1}+1)+(f_{k+1}+1)}{2}$ and $f_0 = f_n = 0$. Then $f_{n-1} = (n-1)$. So expected time to visit all the states is $t_n = t_{n-1} + (n-1) = t_{n-2} + (n-2) + (n-1) = \dots = t_0 + \frac{n(n-1)}{2} = \frac{n(n-1)}{2}$ as $t_0 = 0$.

4.2.2 n-complete graph:

Hitting Time: For a complete graph $t_{hit} = \max_{x,y \in \chi} E_x(\tau_y) = E_x(\tau_y) \forall x, y \in \chi$.

Now here, $\deg(x) = (n-1)$ for all $x \in \chi$.

Now to go from any vertex x to other vertex y in k steps, we go through a path $x - z_1 - z_2 - \dots - z_{k-1} - y$, we choose z_i 's such that $z_i \neq y, \forall i = 1(1)(k-1)$, so basically we don't choose the edge which leads to y in first $(k-1)$ steps and choose the edge which leads to y in the k th step, probability to go from any vertex x to other vertex y in k steps equals $k \cdot \frac{(n-2)^{k-1}}{(n-1)^k}$. Hence,

$$\begin{aligned} E_x(\tau_y) &= 1 \cdot \frac{1}{n-1} + 2 \cdot \frac{1}{n-1} \frac{n-2}{n-1} + 3 \cdot \frac{1}{n-1} \frac{n-2}{n-1} \frac{n-2}{n-1} + \dots + k \cdot \frac{(n-2)^{k-1}}{(n-1)^k} + \dots \\ &= \sum_{k=1}^{\infty} k \cdot \frac{(n-2)^{k-1}}{(n-1)^k} \\ &= \frac{1}{n-1} \sum_{k=1}^{\infty} k \cdot \left(\frac{n-2}{n-1}\right)^{k-1} \\ &= \frac{(n-1)^2}{n-1} = (n-1). \end{aligned}$$

So, $t_{hit} = (n-1)$.

Cover Time For Complete Graph without Loops:

By Mathew's Method, $t_{cov} \leq (n-1)(1 + \frac{1}{2} + \dots + \frac{1}{n-1})$ and $t_{cov} \geq \max_{A \subseteq \chi} (n-1)(1 + \frac{1}{2} + \dots + \frac{1}{|A|-1}) = (n-1)(1 + \frac{1}{2} + \dots + \frac{1}{n-1})$, hence $t_{cov} = (n-1)(1 + \frac{1}{2} + \dots + \frac{1}{n-1}) = \Theta(n \log n)$.

Cover Time For Complete Graph with Loops: Consider a n-complete graph with loops. Then, the expected time when all the states are visited, is equivalent to

the expected time when all types of coupon in a coupon-collector problem.
So the cover time

$$\begin{aligned}
 t_{cov} &= E(\tau_{cov}) \\
 &= \sum_{i=1}^n E(t_i) (\text{where } t_i' s \sim Geo(p_i) \ni p_i = \frac{n-i+1}{n}) \\
 &= \sum_{i=1}^n \frac{n}{n-i+1} \\
 &= \Theta(n \log n).
 \end{aligned}$$

Removing the loops doesn't affect the cover-time too much and hence for n -complete graph without loops, the cover time is $O(n \log n)$.

4.2.3 $(n_1 + n_2)$ complete bipartite graph:

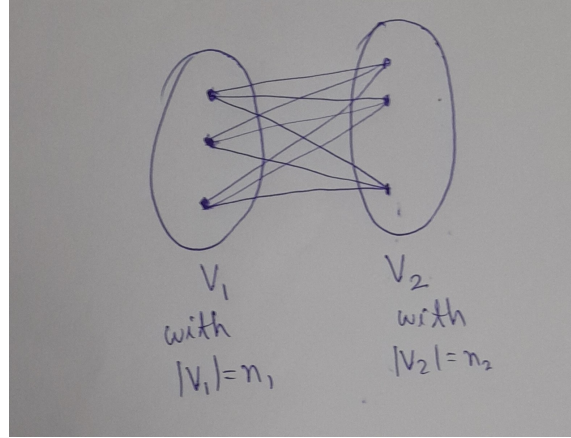


Figure 10: $(n_1 + n_2)$ complete bipartite graph

Hitting Time: Consider a $(n_1 + n_2)$ -complete bipartite graph.

Case-1: $x \in V_1, y \in V_2$

$$\begin{aligned}
\mathbb{E}_x(\tau_y) &= 1 \cdot \frac{1}{n_2 - 1} + 3 \cdot \frac{n_2 - 2}{n_2 - 1} \cdot \frac{1}{n_2 - 1} + 5 \cdot \frac{n_2 - 2}{n_2 - 1} \cdot \frac{n_2 - 2}{n_2 - 1} \cdot \frac{1}{n_2 - 1} + \dots \\
&= \sum_{k=1}^{\infty} \frac{(2k-1)(n_2-2)^{k-1}}{(n_2-1)^k} \\
&= \frac{2}{n_2-1} \sum_{k=1}^{\infty} k \frac{(n_2-2)^{k-1}}{(n_2-1)^{k-1}} - \frac{1}{n_2-1} \sum_{k=1}^{\infty} \frac{(n_2-2)^{k-1}}{(n_2-1)^{k-1}} \\
&= 2(n_2-1) - 1 \\
&= 2n_2 - 3.
\end{aligned} \tag{1}$$

Case-2: $x \in V_2, y \in V_1$

Similarly as case-1, we get $\mathbb{E}_x(\tau_y) = 2n_1 - 3$.

Case-3: $x \in V_1, y \in V_1$

$$\begin{aligned}
\mathbb{E}_x(\tau_y) &= 2 \cdot \frac{1}{n_1 - 1} + 4 \cdot \frac{n_1 - 2}{n_1 - 1} \cdot \frac{1}{n_1 - 1} + \dots \\
&= \frac{2}{n_1 - 1} \sum_{k=1}^{\infty} k \frac{(n_1 - 2)^{k-1}}{(n_1 - 1)^{k-1}} \\
&= \frac{2}{n_1 - 1} (n_1 - 1)^2 \\
&= 2(n_1 - 1).
\end{aligned} \tag{2}$$

Case-4: $x \in V_2, y \in V_2$

Similarly as case-3, we get $\mathbb{E}_x(\tau_y) = 2(n_2 - 1)$.

Now $2(n_i - 1) > 2n_i - 3, \forall i = 1, 2$.

So, $t_{hit} = \max_{x,y \in \chi} \mathbb{E}_x(\tau_y) = 2(\max\{n_1, n_2\} - 1)$.

Cover Time:

By Mathew's Method, $t_{cov} \leq 2(\max\{n_1, n_2\} - 1)(1 + \frac{1}{2} + \dots + \frac{1}{(n_1+n_2-1)})$ and $t_{cov} \geq \max_{A \subseteq \chi} 2(\max\{n_1, n_2\} - 1)(1 + \frac{1}{2} + \dots + \frac{1}{|A|-1}) = t_{cov} \leq 2(\max\{n_1, n_2\} - 1)(1 + \frac{1}{2} + \dots + \frac{1}{(n_1+n_2)})$. So, $t_{cov} = \Theta(\max\{n_1, n_2\} \log(n_1 + n_2))$.

5 Random Walk on Tree:

5.1 Speed of Random Walk on Tree:

Consider a simple random walk on a infinite tree T starting from root o . The random walk moves from a vertex x to one of it's neighbours with probability $\frac{1}{\deg(x)}$. Define a quantity **push away from the root** by function f , where $f(x)$ measures the expected change in the distance from the root in the next step, given that the walk is currently at x .

Denote the distance of the random walk from it's root at time n as $|X_n|$. Suppose the random walk is at $x \neq o$. Now among $\deg(x)$ neighbours of x , one is it's parent which we can reach by taking step (-1) with probability $\frac{1}{\deg(x)}$ and remaining $(\deg(x) - 1)$ are it's children each of which we can reach by taking step $(+1)$ with probability $\frac{1}{\deg(x)}$.

Then,

$$\begin{aligned} f(x) &= \mathbb{E}(|X_{n+1}| - |X_n| | X_n = x) \\ &= \frac{(\deg(x) - 1)}{\deg(x)} \cdot (+1) + \frac{1}{\deg(x)} \cdot (-1) \\ &= \frac{(\deg(x) - 2)}{\deg(x)}. \end{aligned}$$

Now if $x = o$, then the walk can move away from the root to one of it's child by taking step 1 with probability $\frac{1}{\deg(x)}$ for each child. So, $f(x) = \frac{1}{\deg(x)} \cdot \deg(x) = 1$.

$$\text{Hence, } f(x) = \begin{cases} \frac{\deg(x)-2}{\deg(x)} & \text{if } x \neq o \\ 1 & \text{if } x = o \end{cases}.$$

So, we can take $(|X_{n+1}| - |X_n| - f(X_n))_{n \geq 1}$ as a martingale difference sequence as $\mathbb{E}(|X_{n+1}| - |X_n| - f(X_n) | \mathcal{F}_n) = 0$, where $(\mathcal{F}_n)_{n \geq 1}$ is filtration of σ -algebras (i.e; $\mathcal{F}_t \subseteq \mathcal{F}_s, \forall t \leq s$) and X_n is adapted to $\mathcal{F}_n$ (i.e; X_n is $\mathcal{F}_n$ -measurable).

Now as $(|X_{n+1}| - |X_n| - f(X_n))_{n \geq 1}$ is a martingale difference sequence, it is uncorrelated by [A.2](#). Then by SLLN,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n (|X_k| - |X_{k-1}| - f(X_{k-1})) = \lim_{n \rightarrow \infty} \frac{1}{n} (|X_n| - \sum_{k=1}^{n-1} f(X_k)) = 0 \quad \text{a.s.}$$

Now as the density of times at which the random walk visits the root is 0 (since the tree is infinite and has a stationary measure to it's degree and by A.3). Hence $f(X_k) = \frac{\deg(X_k)-2}{\deg(X_k)}$ a.s. which implies $\lim_{n \rightarrow \infty} \frac{1}{n}(|X_n| - \sum_{k=1}^{n-1} (1 - \frac{2}{\deg(X_k)})) = 0$ a.s. which interprets that the speed of the random walk at x is almost surely $\frac{\deg(x)-2}{\deg(x)}$. For a d -ary tree, it becomes almost surely $\frac{d-2}{d}$ for random walk at any state.

A Appendix:

A.1 Polya's Urn Process:

Start with an urn containing two balls, one black and one white. From this point on, proceed by choosing a ball at random from those already in the urn; return the chosen ball to the urn and add another ball of the same colour. If there are j black balls in the urn after k balls have been added (so that there are $k + 2$ balls total in the urn), then the probability that another black ball is added is $\frac{j}{k+2}$. The sequence of ordered pairs listing the numbers of black and white balls is a Markov chain with state space $\{1, 2, \dots\}^2$.

Polya's Urn Lemma: Let B_k be the number of black balls in Polya's urn after the addition of k balls. The distribution of B_k is uniform on $\{1, 2, \dots, k + 1\}$.

A.2 Martingale Difference Sequence Is Uncorrelated:

Suppose $(M_n)_{n \geq 1}$ be a martingale sequence w.r.t. filtration $(\mathcal{F}_n)_{n \geq 1}$. Consider $(M_{j+1} - M_j), (M_i - M_{i-1})$ for $j > i$.

$$Cov(M_{j+1} - M_j, M_i - M_{i-1}) = \mathbb{E}((M_{j+1} - M_j)(M_i - M_{i-1})) - \mathbb{E}(M_{j+1} - M_j)\mathbb{E}(M_i - M_{i-1}).$$

Now for all n ,

$$\begin{aligned} \mathbb{E}(M_{n+1} - M_n) &= \mathbb{E}(\mathbb{E}(M_{n+1} - M_n | \mathcal{F}_n)) \\ &= \mathbb{E}(\mathbb{E}(M_{n+1} | \mathcal{F}_n) - M_n) \quad (\text{as } M_n \text{ adapted to } \mathcal{F}_n) \\ &= \mathbb{E}(0) \quad (\text{by definition of martingale, } \mathbb{E}(M_{n+1} | \mathcal{F}_n) = M_n) \\ &= 0. \end{aligned}$$

So we get, $Cov(M_{j+1} - M_j, M_i - M_{i-1}) = \mathbb{E}((M_{j+1} - M_j)(M_i - M_{i-1}))$. Now,

$$\begin{aligned}\mathbb{E}((M_{j+1} - M_j)(M_i - M_{i-1})) &= \mathbb{E}(\mathbb{E}((M_{j+1} - M_j)(M_i - M_{i-1})|\mathcal{F}_i)) \\ &= \mathbb{E}((M_{j+1} - M_j)\mathbb{E}(M_i - M_{i-1}|\mathcal{F}_i)) \\ &\text{(as } (M_{j+1} - M_j) \text{ adapted to } \mathcal{F}_i) \\ &= \mathbb{E}((M_{j+1} - M_j).0) \quad \text{(by definition of martingales)} \\ &= 0.\end{aligned}$$

Hence we get $Cov(M_{j+1} - M_j, M_i - M_{i-1}) = 0$ and this implies martingale difference sequences are uncorrelated.

A.3 A Result On Expected Return time Of Random Walk on Networks

Theorem A.1. *Let G be a network with $\sum_{e \in E} c(e) = \infty$. Then for all vertex $a \in G$, the expected time for a random walk started at a to return to a is ∞ .*

Proof. For random walk on network with edge-weight $c(x, y)$ for any edge $(x, y) \in E$, we have $\pi(z) = \frac{\sum_{x|z \sim x} c(z, x)}{\sum_{e \in E} c(e)}$ as it's stationary distribution. Now $\sum_{e \in E} c(e) = \infty$ implies $\pi(z) = 0 \quad \forall z$.

Hence there is no stationary distribution which implies that it is null recurrent or transient and hence $\mathbb{E}_z(\tau_z^+) = \infty \quad \forall z$. $\square$